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Day-1: 01.09.2026

Deadline: 01.09.2026

Part A — User and System Investigation (10 minutes)

Determine the username of the account you are currently using.

Display your UID, primary GID, and supplementary groups.

Display all users currently logged in to the system.

Determine what activities the currently logged-in users are performing.

Examine /etc/passwd and identify the following information for your current account:

Username

UID

GID

Home directory

Login shell

Determine how many user-account entries are present in /etc/passwd.

Display only the username and UID fields from /etc/passwd. Find all accounts whose UID is 0.

Explain the security significance of UID 0. Compare /etc/passwd and /etc/shadow. Why are password hashes stored in /etc/shadow instead of /etc/passwd?

```
siva@linux: ~
siva@linux: ~/linuxOS-lab-theory

siva@linux:~/linuxOS-lab-theory$ cd ..
siva@linux:~/linuxOS-lab-theory$ cd ~
siva@linux:~$ whoami
siva
siva@linux:~$ id
uid=1000(siva) gid=1000(siva) groups=1000(siva),27(sudo)
siva@linux:~$ id -u
1000
siva@linux:~$ id -g
1000
siva@linux:~$ id -ng
siva
siva@linux:~$ who
siva
siva@linux:~$ w
 04:24:37 up 33 min,  1 user,  load average: 1.00, 0.63, 0.25
USER          TTY      FROM             LOGIN@   IDLE   JCPU   PCPU   WHAT
siva          tty2    -                03:52   32:57   0.04s  0.04s  /usr/libexec/gnome-session-init-worker ubuntu
siva@linux:~$ grep "^$(whoami):" /etc/passwd
siva:x:1000:1000:siva:/home/siva:/bin/bash
siva@linux:~$ wc -l /etc/passwd
50 /etc/passwd
siva@linux:~$ cut -d: -f1,3 /etc/passwd
root:0
daemon:1
bin:2
sys:3
sync:4
games:5
man:6
lp:7
mail:8
news:9
uucp:10
proxy:13
www-data:33
backup:34
list:38
```

```
siva@linux: ~  
siva@linux: ~/linuxOS-lab-theory  
siva@linux:~/linuxOS-lab-theory$ sudo cat /etc/passwd  
mysql:x:997:1001::/home/mysql:/bin/sh  
sivaranakrishnan:x:1002:1002:sivaranakrishnan,,,:/home/sivaranakrishnan:/bin/bash  
siva@linux:~$ sudo cat /etc/shadow  
[[sudo: authenticate] Password:  
root:*:20566:0:99999:7:::  
daemon:*:20566:0:99999:7:::  
bin:*:20566:0:99999:7:::  
sys:*:20566:0:99999:7:::  
sync:*:20566:0:99999:7:::  
games:*:20566:0:99999:7:::  
man:*:20566:0:99999:7:::  
lp:*:20566:0:99999:7:::  
mail:*:20566:0:99999:7:::  
news:*:20566:0:99999:7:::  
uucp:*:20566:0:99999:7:::  
proxy:*:20566:0:99999:7:::  
www-data:*:20566:0:99999:7:::  
backup:*:20566:0:99999:7:::  
list:*:20566:0:99999:7:::  
irc:*:20566:0:99999:7:::  
_apt:*:20566:0:99999:7:::  
nobody:*:20566:0:99999:7:::  
systemd-networkd:*:20566:1:::  
dhcpcd:*:20566:1:::  
messagebus:*:20566:1:::  
syslog:*:20566:1:::  
systemd-resolve:*:20566:1:::  
_chrony:*:20566:1:::  
tss:*:20566:1:::  
uidd:*:20566:1:::  
systemd-oom:*:20566:1:::  
whoopsie:*:20566:1:::  
dnsmasq:*:20566:1:::  
avahi:*:20566:1:::  
nm-openvpn:*:20566:1:::  
tcpdump:*:20566:1:::  
sssd:*:20566:1:::  
speech-dispatcher:*:20566:1:::  
usbmux:*:20566:1:::  
cups-pk-helper:*:20566:1:::  
fwupd-refresh:*:20566:1:::  
saned:*:20566:1:::  
geoclue:*:20566:1:::  
cups-browsed:*:20566:1:::  
pipewire:*:20566:1:::  
hplip:*:20566:1:::  
gnome-remote-desktop:*:20566:1:::  
polkitd:*:20566:1:::  
rtkit:*:20566:1:::  
colord:*:20566:1:::  
gdm:*:20566:1:::  
siva:$6$1d141U0AvdL2ZcMo$wchtvHeF/xQqwjwGnRG9YX45PNv2fUtxjITj4V1w448Hqu9cR0a3igeREy5nqLuJLC8By3PwXih1BjAwLFJL  
mysql:*:20684:1:::  
sivaranakrishnan:$y$5j9T5Hn6ATHPOSSBR5asCOG/FP/$HkEO2H49YbTfPkf.Twg.g4UyMpIrSXksG3nZVC6Y4n5:20693:0:99999:7:::  
siva@linux:~$ ls -l /etc/passwd /etc/shadow  
-rw-r--r-- 1 root root 2086 Aug 28 13:58 /etc/passwd  
-rw-r----- 1 root shadow 1440 Aug 28 13:57 /etc/shadow
```

```
siva@linux: ~  
siva@linux: ~/linuxOS-lab-theory  
nobody:*:20566:0:99999:7:::  
systemd-networkd:*:20566:1:::  
dhcpcd:*:20566:1:::  
messagebus:*:20566:1:::  
syslog:*:20566:1:::  
systemd-resolve:*:20566:1:::  
_chrony:*:20566:1:::  
tss:*:20566:1:::  
uidd:*:20566:1:::  
systemd-oom:*:20566:1:::  
whoopsie:*:20566:1:::  
dnsmasq:*:20566:1:::  
avahi:*:20566:1:::  
nm-openvpn:*:20566:1:::  
tcpdump:*:20566:1:::  
sssd:*:20566:1:::  
speech-dispatcher:*:20566:1:::  
usbmux:*:20566:1:::  
cups-pk-helper:*:20566:1:::  
fwupd-refresh:*:20566:1:::  
saned:*:20566:1:::  
geoclue:*:20566:1:::  
cups-browsed:*:20566:1:::  
pipewire:*:20566:1:::  
hplip:*:20566:1:::  
gnome-remote-desktop:*:20566:1:::  
polkitd:*:20566:1:::  
rtkit:*:20566:1:::  
colord:*:20566:1:::  
gdm:*:20566:1:::  
siva:$6$1d141U0AvdL2ZcMo$wchtvHeF/xQqwjwGnRG9YX45PNv2fUtxjITj4V1w448Hqu9cR0a3igeREy5nqLuJLC8By3PwXih1BjAwLFJL  
mysql:*:20684:1:::  
sivaranakrishnan:$y$5j9T5Hn6ATHPOSSBR5asCOG/FP/$HkEO2H49YbTfPkf.Twg.g4UyMpIrSXksG3nZVC6Y4n5:20693:0:99999:7:::  
siva@linux:~$ ls -l /etc/passwd /etc/shadow  
-rw-r--r-- 1 root root 2086 Aug 28 13:58 /etc/passwd  
-rw-r----- 1 root shadow 1440 Aug 28 13:57 /etc/shadow
```

Part B — Creating and Verifying Users (20 Minutes)

Create the following users and ensure that a home directory is created for each:

alice, bob, charlie

Verify that all three accounts have been added to `/etc/passwd`.

Determine the UID assigned to each newly created user.

Determine the home directory and default shell assigned to alice.

Inspect `/home` and verify that the required home directories were created.

Set passwords for all three accounts.

Switch from your current account to alice using a login shell. Verify the change using an appropriate command, and then return to your original account.

```
siva@linux: ~ --sh
siva@linux: ~/linuxOS-lab-theory
-rw-r--r-- 1 root root 2000 Aug 28 13:28 /etc/passwd
-rw-r--r-- 1 root shadow 1440 Aug 28 13:57 /etc/shadow
siva@linux:~$ sudo useradd -m alice
siva@linux:~$ ^C
siva@linux:~$ sudo useradd -m alice
useradd: user 'alice' already exists
siva@linux:~$ sivalin@123
sivalin@123: command not found
siva@linux:~$ sudo useradd -m alice
useradd: user 'alice' already exists
siva@linux:~$ sudo useradd -m charlie
siva@linux:~$ sudo useradd -m charlie
useradd: user 'charlie' already exists
siva@linux:~$ sudo useradd -m bob
siva@linux:~$ sudo useradd -m bob
useradd: user 'bob' already exists
siva@linux:~$ grep -E '^(alice|bob|charlie):' /etc/passwd
alice:x:1003:1003::/home/alice:/bin/sh
charlie:x:1004:1004::/home/charlie:/bin/sh
bob:x:1005:1005::/home/bob:/bin/sh
siva@linux:~$ id -u alice
1003
siva@linux:~$ id -u bob
1005
siva@linux:~$ id -u charlie
1004
siva@linux:~$ getent passwd alice
alice:x:1003:1003::/home/alice:/bin/sh
siva@linux:~$ echo "Home: $(getent passwd alice | cut -d: -f6)"
Home: /home/alice
siva@linux:~$ echo "Shell: $(getent passwd alice | cut -d: -f7)"
Shell: /bin/sh
siva@linux:~$ ls -l /home
total 20
drwxr-x--- 2 alice      alice      4096 Sep  1 04:32 alice
drwxr-x--- 2 bob        bob        4096 Sep  1 04:36 bob
drwxr-x--- 2 charlie    charlie    4096 Sep  1 04:34 charlie
drwxr-x--- 19 siva      siva       4096 Aug 28 15:20 siva
```

```
siva@linux: ~ -- -sh

siva@linux: ~/linuxOS-lab-theory

siva@linux:~$ ls -ld /home/alice /home/bob /home/charlie
drwxr-x--- 2 alice  alice  4096 Sep  1 04:32 /home/alice
drwxr-x--- 2 bob   bob    4096 Sep  1 04:36 /home/bob
drwxr-x--- 2 charlie charlie 4096 Sep  1 04:34 /home/charlie
siva@linux:~$ sudo passwd alice
New password:
BAD PASSWORD: The password contains the user name in some form
Retype new password:
Sorry, passwords do not match.
New password:
BAD PASSWORD: The password contains the user name in some form
Retype new password:
Sorry, passwords do not match.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
siva@linux:~$ sudo passwd bob
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
siva@linux:~$ sudo passwd charlie
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
siva@linux:~$ su -alice
su: invalid option -- 'a'
Try 'su --help' for more information.
siva@linux:~$ su alice
Password:
$
$ whoami
alice
$ pwd
/home/siva
```

```
siva@linux: ~ -- -sh

siva@linux: ~/linuxOS-lab-theory

siva@linux:~$ cd ..
siva@linux:/home$ ls -ld /home/alice /home/bob /home/charlie
drwxr-x--- 2 alice  alice  4096 Sep  1 04:48 /home/alice
drwxr-x--- 2 bob   bob    4096 Sep  1 04:36 /home/bob
drwxr-x--- 2 charlie charlie 4096 Sep  1 04:34 /home/charlie
siva@linux:/home$ su alice
Password:
$ pwd
/home
$ whoami
alice
$ pwd
/home
$ exit
siva@linux:/home$ cd -
siva@linux:~$ cd -
siva@linux:~$ su alice
Password:
$ pwd
/home/siva
$ whoami
alice
$ exit
siva@linux:~$ getent passwd alice
alice:x:1003:1003::/home/alice:/bin/sh
siva@linux:~$ su alice
Password:
$ pwd
/home/siva
$ sudo usermod -d /home/alice alice
sudo: I'm sorry alice, I'm afraid I can't do that
$ exit
siva@linux:~$ sudo usermod -d /home/alice alice
usermod: no changes
siva@linux:~$ ls -ld /home/alice
drwxr-x--- 2 alice alice 4096 Sep  1 04:48 /home/alice
siva@linux:~$ sudo mkdir /home/alice
```

Part C — Group Management and Access Control (30 Minutes)

Create two groups:

security and developers

Add alice and charlie to the security group.

Add bob to the developers group.

Without switching accounts, verify the group membership of all three users.

Display the entries corresponding to security and developers from /etc/group.

Add bob to the security group without removing him from the developers group.

Verify that bob now belongs to both groups.

```
siva@linux: ~  
siva@linux: ~/linuxOS-lab-theory  
$ echo $SHELL  
/bin/sh  
$ exit  
siva@linux:~$ sudo groupadd security  
siva@linux:~$ sudo groupadd developers  
siva@linux:~$ getent group security  
security:x:1006:  
siva@linux:~$ getent group developers  
developers:x:1007:  
siva@linux:~$ sudo usermod -aG security alice  
siva@linux:~$ sudo usermod -aG developers bob  
siva@linux:~$ groups alice  
alice : alice security  
siva@linux:~$ groups bob  
bob : bob developers  
siva@linux:~$ groups charlie  
charlie : charlie  
siva@linux:~$ sudo usermod -aG security charlie  
siva@linux:~$ groups charlie  
charlie : charlie security  
siva@linux:~$ id alice  
uid=1003(alice) gid=1003(alice) groups=1003(alice),1006(security)  
siva@linux:~$ id bob  
uid=1005(bob) gid=1005(bob) groups=1005(bob),1007(developers)  
siva@linux:~$ id charlie  
uid=1004(charlie) gid=1004(charlie) groups=1004(charlie),1006(security)  
siva@linux:~$ grep -E '^(security|developers):' /etc/group  
security:x:1006:alice,charlie  
developers:x:1007:bob  
siva@linux:~$ groupname : password : GID : members  
groupname: command not found  
siva@linux:~$ sudo usermod -aG security bob  
siva@linux:~$ groups bob  
bob : bob security developers  
siva@linux:~$ id bob  
uid=1005(bob) gid=1005(bob) groups=1005(bob),1006(security),1007(developers)  
siva@linux:~$
```